G. Sai praneeth

([redacted])

* 2 stage OTA design Assignment:

Given $V_{Tn} = 0.37V$; $V_{TP} = 0.39V$

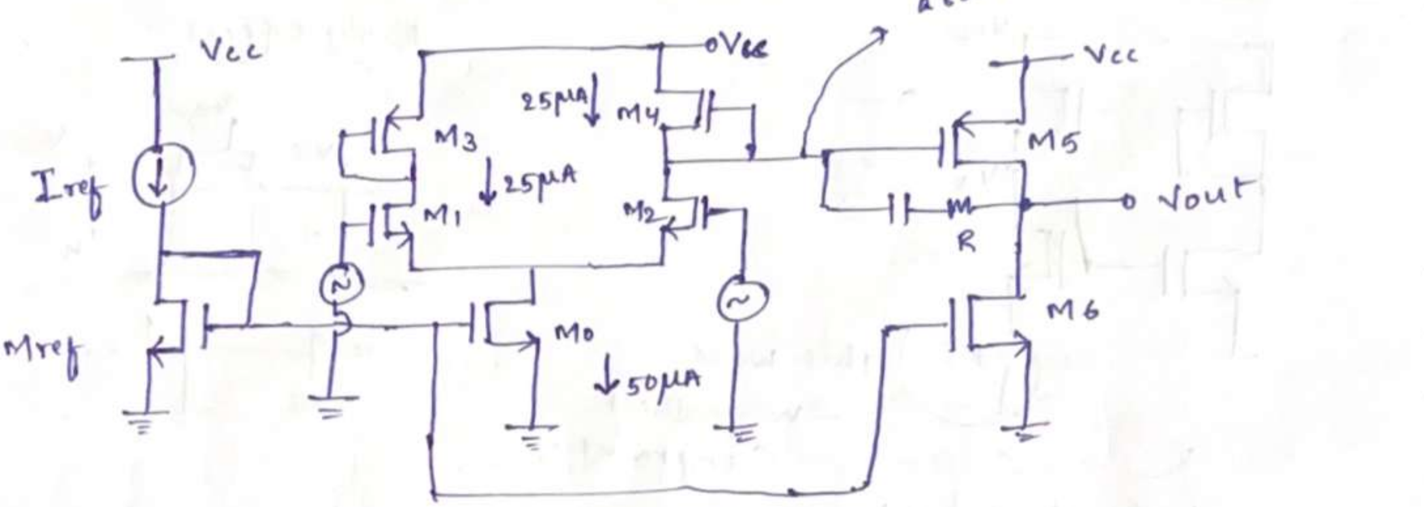
$\mu_n C_{ox} = 230 \mu A/V^2$; $\mu_p C_{ox} = 100 \mu A/V^2$

$V_{dd} = 0.18 \times 10 \, V$; $L_{min} = 0.18 \mu m$; $W_{min} = 0.27 \mu m$.

* Choose tail current $= 50 \mu A$, $I_{ref} = 5 \mu A$.

Current through $M_1, M_2, M_3, M_4 = 25 \mu A$.

By assuming overdrive for M_1, M_2 is $100 \, mV = 0.1V$



$$I_D \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2$$

$\mu_n C_{ox} = 230 \, A/V^2$

$$25 \mu A = \frac{1}{2} (230 \times 10^{-6}) \left(\frac{W}{L}\right) (0.1)^2$$

$\boxed{\frac{W}{L} \approx 21.74}$ * Choose $\boxed{L_1 = 1 \mu m, \; W_1 = 21.74 \mu m}$

* Since same current is flowing through M_2:

$\boxed{L_2 = 1 \mu m, \; W_1 = 21.74 \mu m.}$

* for Better gain } Mref $\Rightarrow$ $\boxed{L = 1 \mu m, \; W = 10 \mu m}$

* Since $\boxed{I_{tail} = 10 \times I_{ref}}$ $\Rightarrow$ $\boxed{L_0 = 1 \mu m, \; W_0 = 100 \mu m}$

for $M_3 \Rightarrow \quad I_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)(V_{gs}-V_{th})^2$

$$2\times 25\times 10^{-6} = 100\times 10^{-6} \times \frac{W}{L} \times (V_{sg}-V_{th})^2$$

for pmos, $V_{sg}-V_{th} = V_{cc} - V_G - (V_{tp})_{pmos}$

$$= 1.8V - 1V - 0.39V$$

$$= 0.41V$$

$$50\times 10^{-6} = 100\times 10^{-6} \times \frac{W}{L} \times (0.41)^2$$

$$\boxed{\frac{W}{L} \approx 2.974}$$

* Choose $\boxed{\begin{array}{l} L_3 = 1\mu m \\ W_3 = 2.974 m \end{array}}$

* Since same current passing through M_4

$$\boxed{L_4 = 1\mu m, W_4 = 2.974\mu m}$$

* As we know the current passing through $\boxed{I_6 = 50\,\mu A = I_5}$

so $\left(\frac{W}{L}\right)_5 = 2\left(\frac{W}{L}\right)_3$

$$\boxed{L_5 = 1\mu m, \; W_5 = 5.948\mu m}$$

$$\boxed{L_6 = 1\mu m, \; W_6 = 1001\,\mu m}$$ By taking $\underline{V_{gs}-V_{th} = 0.1V}$

theoretical $g_{m1} = \frac{2I_1}{V_{gs}-V_{th}} = \frac{2\times 25\times 10^{-6}}{10^{-1}}$

$$\boxed{g_{m1} = 5\times 10^{-4}}$$

simulated $g_{m1} \approx 4.2\times 10^{-4}$

$$\left.\begin{array}{l} r_{o4} = \frac{1}{G_{ds4}} = \frac{1}{1.00\times 10^{-6}} = 10^6 \\ r_{o2} = \frac{1}{G_{ds2}} = \frac{1}{2.99\times 10^{-6}} = 3.3\times 10^5 \end{array}\right\}$$ By simulation we got the values

$$r_{o1} \,||\, r_{o2} = 2.48\times 10^5$$

* gain of stage 1 $= g_{m1} (r_{o1} \| r_{o2})$

$$= 4.2 \times 10^{-4} (2.48 \times 10^{5})$$

$$= 104.16$$

$20 \log(\text{gain}) = 20 \log(104.16) \approx 40\,dB$

* gain of stage 2

$$g_{m5} = \frac{2I_5}{V_{sg} - V_{th}} = \frac{2 \times 50 \times 10^{-6}}{0.582} = 1.71 \times 10^{-4}$$

Simulated $g_{m5} = 1.63 \times 10^{-4}$

$$r_{o5} = \frac{1}{G_{ds5}} = \frac{1}{2.39 \times 10^{-6}} = 4.18 \times 10^{5}$$

$$r_{o6} = \frac{1}{G_{ds6}} = \frac{1}{6.47 \times 10^{-6}} = 1.54 \times 10^{5}$$

By simulation we got these values

$r_{o5} \| r_{o6} = 1.12 \times 10^{5}$

$$\text{gain}_2 = g_{m5} (r_{o5} \| r_{o6})$$

$$\approx (1.63 \times 10^{-4})(1.12 \times 10^{5})$$

$$\approx 18.256$$

net gain $= (\text{gain}\,1)(\text{gain}\,2) = (104.16)(18.256)$

$$= 1901.54$$

$$\boxed{20 \log(\text{gain}) = 20 \log(1901.54) = 65.58\,dB}$$

let us take $C_2 = 5pf$

so by the miller effect (C_2)input side $= (1+A) C_2$

$$= (1 + 18.256)(5pf)$$

$$= 96.3pf$$

where as (C_2)output side $= (1 + C_2)pf$

$$= 6pf.$$

$$\text{pole 1} \approx \frac{1}{R_{1eff}\,(C_2)\text{input}} = \frac{1}{2.48 \times 10^5 \times 96.28\,pF}$$

$$\underline{P_1 = 41.8\,KHz}$$

$$\text{pole 2} = \frac{1}{R_{2eff}(C_2)\,\text{output}} = \frac{1}{1.12 \times 10^5 \times 6 \times 10^{-12}}$$

$$P_2 = \underline{1400\,KHz}$$

	length (μm)	width (μm)
Mref	1	10
M0	1	100
M1	1	21.74
M2	1	21.74
M3	1	2.94
M4	1	2.94
M5	1	5.94
M6	1	100

Mref: $V_{gs} - V_{th} = 0.456 - 0.408 = 0.048$

$V_{ds} = 0.456$ $\boxed{V_{ds} > V_{gs} - V_{th}}$ ⇒ saturation

M0: $V_{gs} - V_{th} = 0.456 - 0.409 = 0.047$

$V_{ds} = 0.316$ $\boxed{V_{ds} > V_{gs} - V_{th}}$ ⇒ saturation

M1: $V_{gs} - V_{th} > V_{ds}$

$0.584 - 0.5 > 0.5$

$\boxed{0.084 > 0.5}$ ⇒ saturation

M2: $V_{gs} - V_{th} = 0.584 - 0.5 = 0.084$

$V_{ds} = 0.484$

$\boxed{V_{ds} > V_{gs} - V_{th}}$ ⇒ saturation

M3: $V_{sg} - V_{tp} < V_{SD}$

$0.897 - 0.417 < 0.974$

$\boxed{0.557 < 0.974} \Rightarrow$ Saturation

M4: $V_{sg} - V_{tp} < V_{SD}$

$\boxed{0.897 - 0.417 < 0.974} \rightarrow$ saturation

M5: $V_{sg} - V_{tp} = 1 - 0.418 = 0.582$

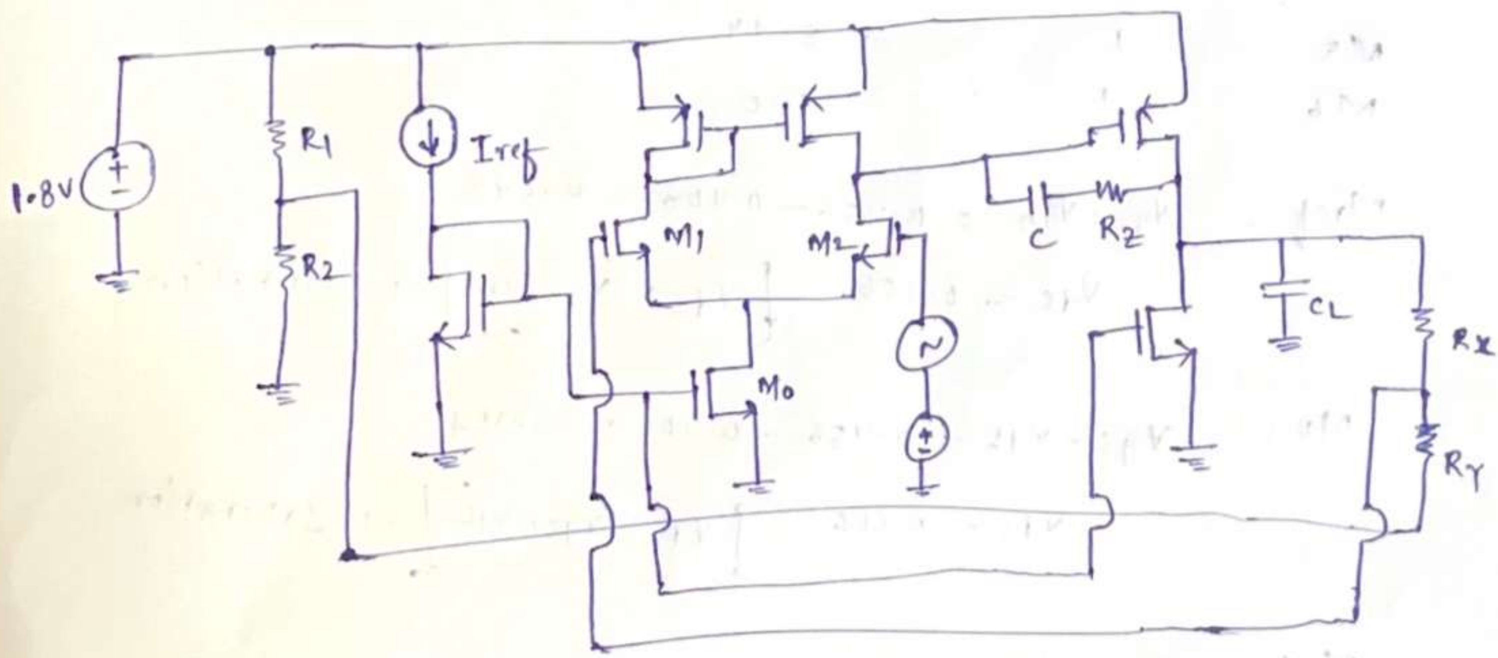
$\boxed{V_{SD} = 0.904} \Rightarrow$ Saturation

M6 = $V_{sg} - V_{th} = 0.456 - 0.405 = 0.05$

$\boxed{V_{ds} = 0.899V} \Rightarrow$ saturation

* all mosfets are in saturation.

* final design of 2stage OTA:-



To get 0.9V we can use voltage divider by adding resistors R1, R2. By calculating poles & zeroes CL taken as

$R_x = R_y = 2000\,K\Omega$ → for closed loop gain

Calculated Bandwidth (seeing when there is drop at 70%) } Closed loop gain = 2

so at 70% ⇒ 20 log(2×0.7) = 2.94 dB } so gain Bandwidth frequency = 888 KHz

→ calculated phase margin by checking phase when magnitude is 0dB.

so phase at 0dB :- −104° (approximately) ⇒ ϕ_{gc}

∴ phase margin = 180° + ϕ_{gc} = 76°

so my system is stable &

$$\boxed{\begin{aligned} C_L &= 13.55\,pF \\ C &= 50.55\,pF \\ R_t &= 7.5\,K\Omega \end{aligned}}$$

* for pulse input:-

$V_{initial} = 0V$

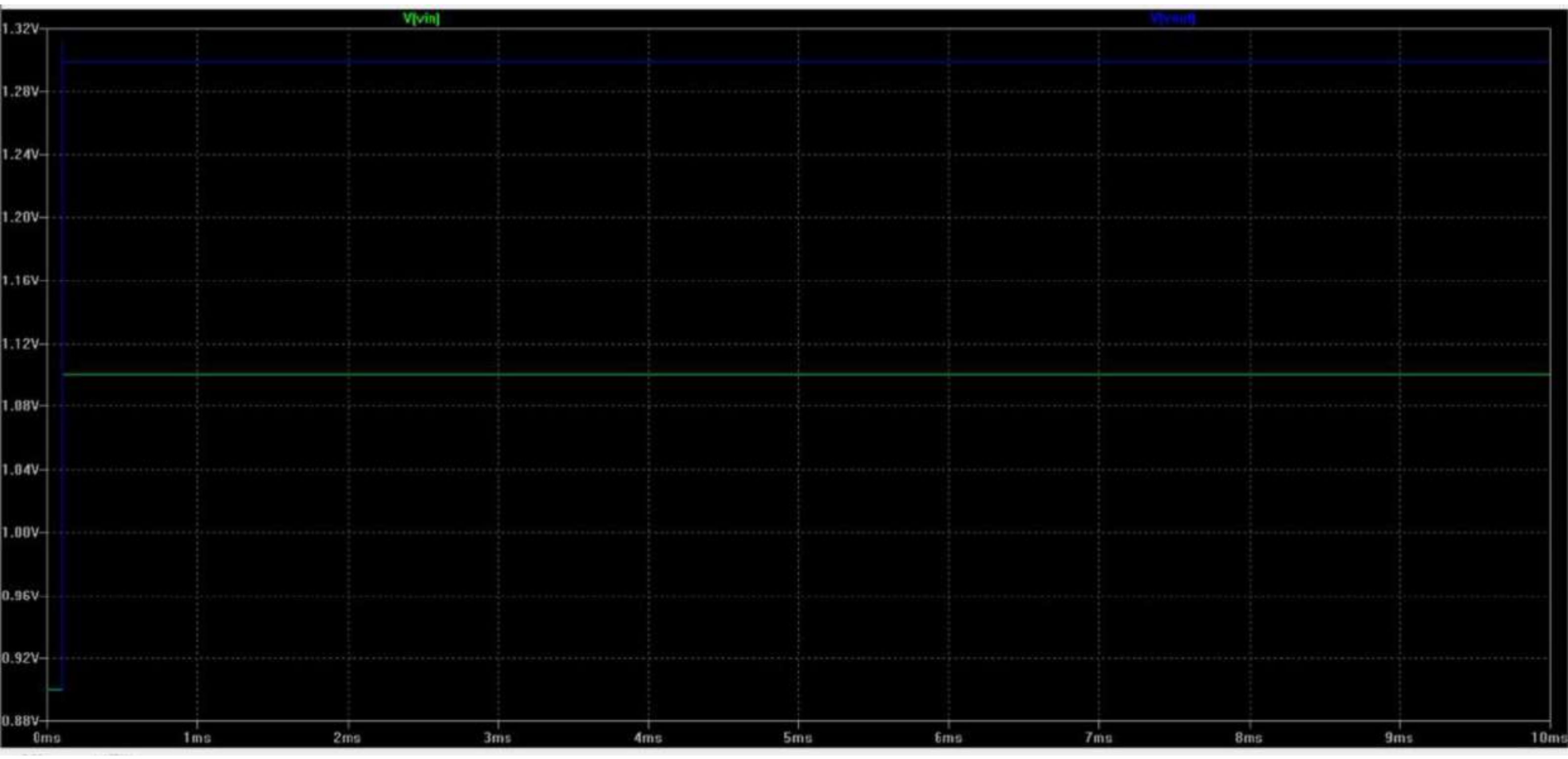
$V_{on} = 0.2\,V$

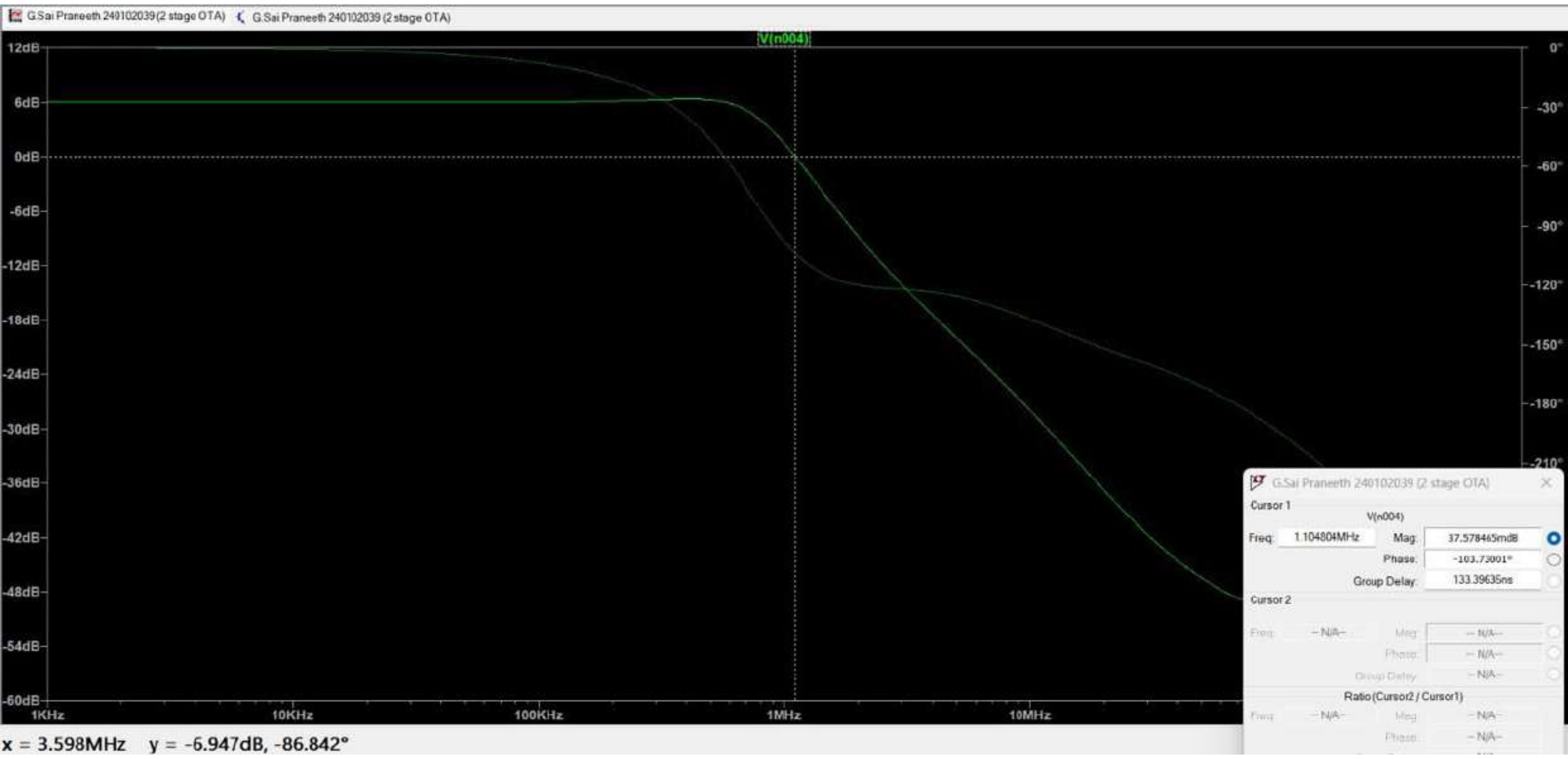
$T_{delay} = 0.1$ msec

$T_{rise} = 0.1$ nsec

$T_{fall} = 0.1$ msec

> The output pulse is :-


V[vin]
1.32V
1.28V
1.24V
1.20V
1.16V
1.12V
1.08V
1.04V
1.00V
0.96V
0.92V
0.88V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms


G.Sai Praneeth 240102039 (2 stage OTA)
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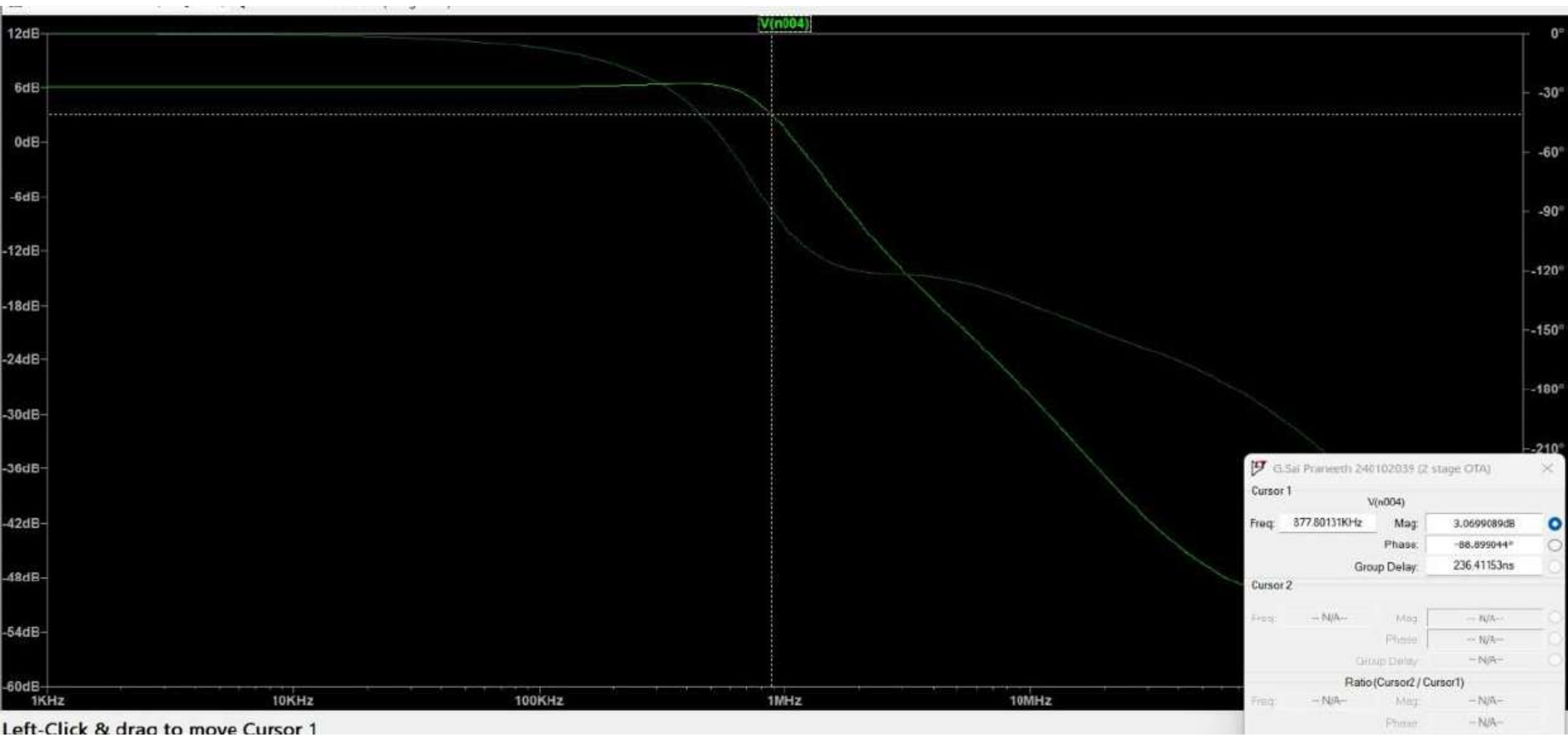
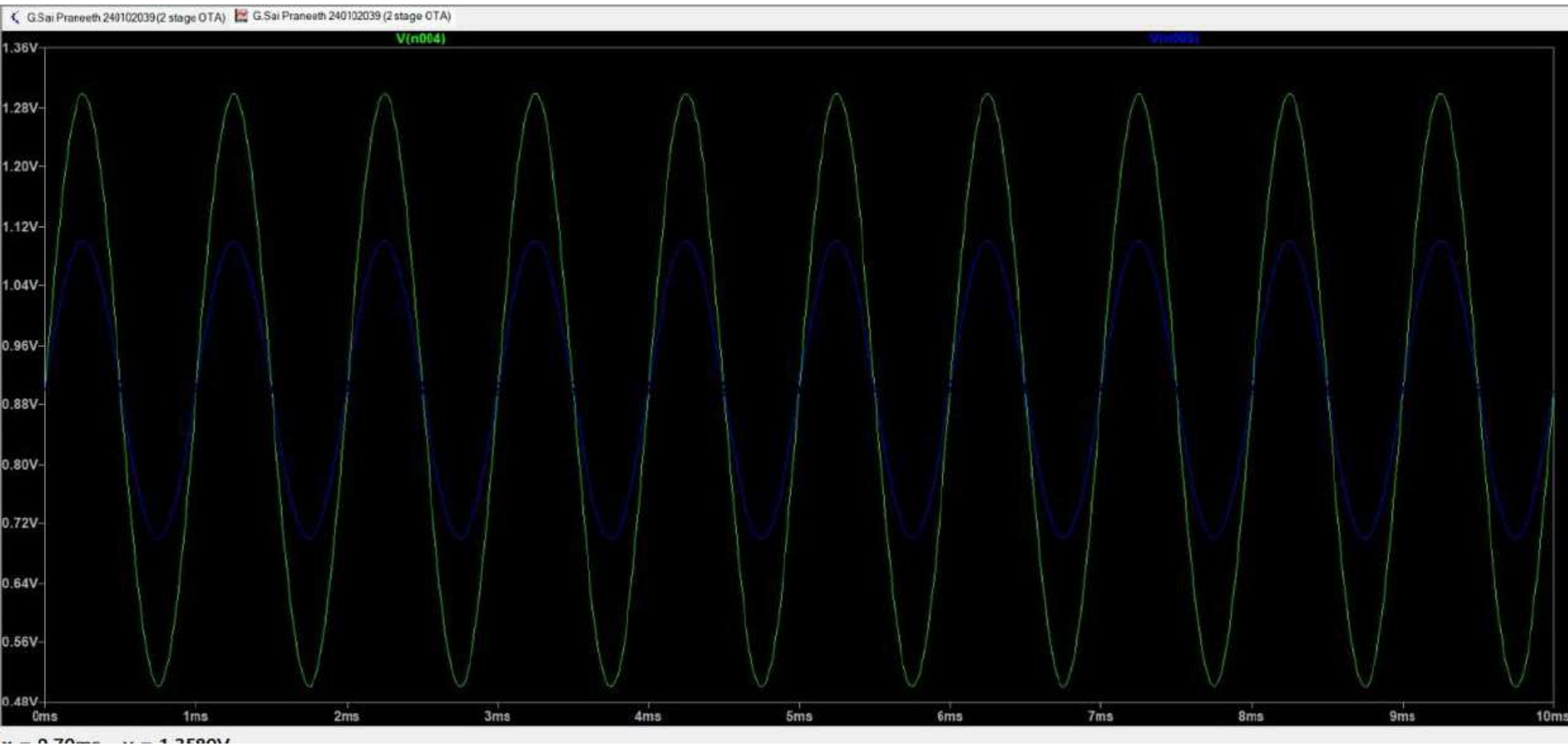
V(n004)
12dB
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
1KHz
10KHz
100KHz
1MHz
10MHz
G.Sai Praneeth 240102039 (2 stage OTA)
Cursor 1
V(n004)
Freq: 1.104804MHz
Mag: 37.578465mdB
Phase: -103.73001°
Group Delay: 133.39635ns
Cursor 2
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
Group Delay: --N/A--
Ratio (Cursor2 / Cursor1)
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
x = 3.598MHz y = -6.947dB, -86.842°


V(n004)
12dB
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
1KHz
10KHz
100KHz
1MHz
10MHz
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
G.Sai Praneeth 240102039 (2 stage OTA)
Cursor 1
V(n004)
Freq: 877.80131KHz
Mag: 3.0699089dB
Phase: -88.899044°
Group Delay: 236.41153ns
Cursor 2
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
Group Delay: --N/A--
Ratio(Cursor2 / Cursor1)
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
Left-Click & drag to move Cursor 1


G.Sai Praneeth 240102039 (2 stage OTA)
G.Sai Praneeth 240102039 (2 stage OTA)
V(n004)
1.36V
1.28V
1.20V
1.12V
1.04V
0.96V
0.88V
0.80V
0.72V
0.64V
0.56V
0.48V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms